



Influenza Virus Testing Methods

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 For Public Health

PURPOSE

Table depicting influenza virus testing methods.

Table: Influenza Virus Testing Methods

Method ^[1]	Types Detected	Acceptable Specimens ^[2]	Test Time	CLIA Waived ^[3]
Rapid Influenza Diagnostic Tests (singleplex and multiplex; antigen detection) ^[4]	A and B	NP swab, aspirate or wash, nasal swab, aspirate or wash, throat swab	<15 min.	Yes/No
Rapid Molecular Assay [influenza viral RNA or nucleic acid detection]	A and B	NP swab, nasal swab	15-30 minutes ^[5]	Yes/No ^[5]
Immunofluorescence, Direct (DFA) or Indirect (IFA) Florescent Antibody Staining [antigen detection]	A and B	NP swab or wash, bronchial wash, nasal or endotracheal aspirate	1-4 hours	No
RT-PCR ^[6] (singleplex and multiplex; real-time and other RNA-based) and other molecular assays [influenza viral RNA or nucleic acid detection]	A and B	NP swab, throat swab, NP or bronchial wash, nasal or endotracheal aspirate, sputum, BAL fluid	Varies (1 to 8 hours, varies by the assay)	No
Rapid cell culture (shell vials; cell mixtures; yields live virus)	A and B	NP swab, throat swab, NP or bronchial wash, nasal or endotracheal aspirate, sputum, BAL fluid (specimens placed in VTM)	1-3 days	No
Viral tissue cell culture (conventional; yields live virus)	A and B	NP swab, throat swab, NP or bronchial wash, nasal or endotracheal aspirate, sputum, BAL fluid (specimens placed in VTM)	3-10 days	No

Abbreviations

- FDA = U.S. Food and Drug Administration
- A = aspirate
- B = bronchial
- BAL = bronchoalveolar lavage
- N = nasal
- NP = nasopharyngeal
- RNA = ribonucleic acid
- S = swab
- T = throat
- TR = tracheal
- UTM = universal transport media
- VTM = viral transport media
- W = wash

SOURCES AND PAGE INFO

CONTENT SOURCE:

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ABOUT THIS PAGE

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The information on this page was last reviewed by subject matter experts to ensure accuracy.

REFERENCES

1. Serologic (antibody detection) testing is not recommended for routine patient diagnosis and cannot inform clinical management. A single acute serum specimen for seasonal influenza serology is uninterpretable and should not be collected. Serological testing for detection of antibodies to seasonal influenza viruses is useful for research studies and requires collection of appropriately timed acute and convalescent serum specimens and testing of paired sera at specialized research or public health laboratories.
2. Approved clinical specimens vary by influenza test. Consult the manufacturer's package insert for the approved clinical specimens for each test. Approved respiratory specimens vary among FDA cleared influenza assays.
3. Clinical Laboratory Improvement Amendments (CLIA) of 1988. <http://www.cms.gov/Regulations-and-Guidance/Legislation/CLIA/index.html>
4. Chromatographic- and/or fluorescence-based lateral flow and membrane-based immunoassays. Some approved rapid influenza diagnostic assays utilize an analyzer reader device.
5. Rapid molecular assays can provide results in approximately 15-30 minutes.
6. Reverse transcription polymerase chain reaction, including U.S. Food and Drug Administration (FDA)-approved test systems, laboratory testing using analyte specific reagents, or laboratory-developed reagents. Some approved molecular assays can produce results in approximately 60-80 minutes.

SOURCES

- Conjunctival swabs are currently approved under FDA enforcement discretion for testing with the CDC Influenza A/H5 subtyping assay. Conjunctival swabs are an acceptable clinical specimen for a commercially available A(H5) assay. More information is available at [Commercial Laboratories: Novel Influenza A Virus Testing | Bird Flu | CDC](#) and [Influenza A \(H5\) Virus | ARUP Laboratories](#).